# 1 Core-Hole Potential

Instead of an explicit calculating of the overlap term, $S_{pq}$ - which depends on the final relaxed orbital $\phi_p$ and initial orbital $\psi_q$, these can also be obtained in terms of the core-hole potential acting on the initial wave function. In first order perturbation theory the overlap is then,

$$S_{pq} = \langle \phi_p | \psi_q \rangle \approx \frac{\langle \psi_p | \hat{v}_{\rm ch} | \psi_q \rangle}{\epsilon_p - \epsilon_q} \tag{1}$$

with $\psi_p$ and $\psi_q$ the orbitals involved in shake-down where $p \neq q$, and $\epsilon$ defines their MO energies. We wish to compute the matrix-element,

$$\langle \psi_p | \hat{v}_{\rm ch} | \psi_q \rangle \approx \int dr \;\; \psi_p^*(\mathbf{r}) v_{ch}(\mathbf{r}) \psi_q(\mathbf{r}), \tag{2}$$

with,

$$v_{ch}(\mathbf{r}) = -\int d\mathbf{r}' \;\; \frac{\rho_c(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}, \tag{3}$$

where $\rho_c$ is the density of the core orbital hole. We can expand each molecular orbital in the AO basis,

$$\psi_p(\mathbf{r}) = \sum_\alpha C_{\alpha p} \chi_\alpha(\mathbf{r}), \tag{4}$$

giving the AO representation of the matrix element

$$\langle \psi_p | \hat{v}_{\rm ch} | \psi_q \rangle \approx \sum_{\alpha,\beta} C_{\alpha p}^* V_{\alpha\beta} C_{\beta q}, \tag{5}$$

where $V_{\alpha\beta}$ is the AO representation of the core-hole potential $v_{ch}$,

$$V_{\alpha\beta} = -\iint d\mathbf{r}\, d\mathbf{r}' \frac{\chi_\alpha^*(\mathbf{r}) \chi_\beta(\mathbf{r}) \rho_c(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}. \tag{6}$$

Expanding the core-orbital $\psi_c$ (in the density $\rho_c$) in an AO basis contribution,

$$\psi_c(\mathbf{r}) = \sum_\gamma C_{\gamma c} \chi_\gamma(\mathbf{r}), \tag{7}$$

we arrive at an expression for the core-hole potential as a regular two-electron integral,

$$V_{\alpha\beta} = -\sum_{\gamma\delta} C_{\gamma c}^* C_{\delta c} (\alpha\beta|\gamma\delta), \tag{8}$$

where,

$$(\alpha\beta|\gamma\delta) = \iint d\mathbf{r}\, d\mathbf{r}' \frac{\chi_\alpha^*(\mathbf{r}) \chi_\beta(\mathbf{r}) \chi_\gamma^*(\mathbf{r}') \chi_\delta(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}. \tag{9}$$

Thus giving the total matrix element in the AO representation,

$$\boxed{\langle \psi_p | \hat{v}_{\rm ch} | \psi_q \rangle \approx -\sum_{\alpha,\beta} \sum_{\gamma,\delta} C_{\alpha p}^* C_{\beta q} C_{\gamma c}^* C_{\delta c} (\alpha\beta|\gamma\delta)} \tag{10}$$

In MO representation this is simply the four-center MO integral

$$\boxed{\langle \psi_p | \hat{v}_{\rm ch} | \psi_q \rangle \approx -(pq|cc)} \tag{11}$$

Thus,

$$S_{pq} \approx \frac{(pq|cc)}{\epsilon_q - \epsilon_p} \tag{12}$$